SETS

* Union of the set

The union of A and B is the set which consist all the elements of A and all the elements of B, the common elements being taken only ones. The symbol '$\cup$' is used to denote the union.

Example :- Set A = ~~2468~~ $\{2, 4, 6, 8\}$

Set B = $\{6, 8, 10, 12\}$

$A \cup B = \{2, 4, 6, 8, 10, 12\}$

* Intersection of sets

The intersection of set A and B is the set of all the elements which are common to both A and B. The symbol '$\cap$' is used to denote intersection.

Example :- Set A = $\{2, 4, 6, 8\}$

Set B = $\{6, 8, 10, 12\}$

$A \cap B = \{6, 8\}$

Q1:- If $A = \{1, 3, 5, 6, 7\}$

$B = \{2, 4, 1, 6, 7\}$

$C = \{3, 7, 1, 8\}$

verify — (i) $A \cup (B \cup C) = (A \cup B) \cup C$

(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(i) $B \cup C = \{1, 2, 3, 4, 6, 7, 8\}$

$A \cup (B \cup C) = \{1, 2, 3, 4, 5, 6, 7, 8\}$ — (i)

$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$

$(A \cup B) \cup C = \{1, 2, 3, 4, 5, 6, 7, 8\}$ — (ii)

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(ii) $(B \cup C) = \{1, 2, 3, 4, 6, 7, 8\}$

$A \cap (B \cup C) = \{1, 3, 6, 7\}$ — (i)

$(A \cap B) = \{1, 6, 7\}$

$(A \cap C) = \{1, 3, 7\}$

$\therefore (A \cap B) \cup (A \cap C) = \{1, 3, 6, 7\}$ — (ii)

Q If $A = \{0, 1, 2, 3, 4, 7\}$; $B = \{3, 5, 4, 8, 0\}$

$C = \{0, 1, 3, 9, 8\}$, verify —

(i) $A \cap (B \cap C) = (A \cap B) \cap C$

(ii) $A \cup (B \cap C) = (A \cap B) \cap (A \cup C)$

Let $f : R \rightarrow R$ be a function defined by $f(x) = 2x + 5$ and $g : R \rightarrow R$ be a function defined by $g(x) = \frac{x-5}{2}$ show that f and g are inversed each other

Sol - $g : R \rightarrow R$

$$g(x) = \frac{x-5}{2}$$

f and g

$$(f \cdot g)(x) = f \cdot (g(x))$$

$$= f\left(\frac{x-5}{2}\right) - x$$

$$= \cancel{2}\left(\frac{x - \cancel{5}}{\cancel{2}}\right) + \cancel{5}$$

$$= x$$

$$(g \cdot f)(x) = g(f(x))$$

$$= g((2x+5)) - x$$

$$= \frac{(\cancel{2}x + \cancel{5}) - \cancel{5}}{\cancel{2}}$$

$$= x$$

(system of
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Q.

Sol

(matrix) Rank and Inverse

module 1	=	5 marks (1)	12 marks (1)
(system of linear) module 2	=	5 mark (1)	9 marks (1)
(logics) module 3	=	5 mark (1)	9 marks (1)
(sets) module 4	=	5 marks (qs)	9 marks (1)
permutations module 5	=	"	"

Invertible function

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A function f from A to b is said to be invertible if there $\underbrace{\text{exist}}_{\exists}$ $g : B \to A$

such that $g \cdot f = I_1$ and $f \cdot g = I_2$

$= 2x + 5$

$) = \frac{x-5}{2}$

Q. Consider the function $f(x) = x^3$ and $g(x) = x^2 + 1$ $\underbrace{\forall x \text{ belongs to } R}_{\forall x \in R}$. Find $g \cdot f$, $f \cdot g$, f^2 & g^2.

Soln - $(f \cdot g)(x) = f \cdot (g(x))$

$= f(x^2+1)$

$= (x^2+1)^3$ Consider $(a+b)^3$

$= x^6 + 1 + 3x^4 + 3x^2 + 1$

$(g \cdot f)(x) = g \cdot (f(x))$

$= g(x^3)^2 + 1$

$= x^6 + 1$

$f^2 = (f \cdot f)(x) = f(f(x))$

$= f(x^3)$

$= (x^3)^3 = x^9$

$g^2 = (g \cdot g)(x) = g(g(x))$

$= g(x^2+1)$

$= (x^2+1)^2 + 1$

$= x^4 - 2a^2 + 2$

Q Let $f: R \to R$ be a function defined by

$$f(x) = \begin{cases} 3x-5, & x > 0 \\ 1-3x, & x \leq 0 \end{cases}$$ then

find $f(0)$, $f(5/3)$, $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(5/3)$, $f^{-1}(3)$

Consider $f(0)$

first consider $3x-5$

$3(0)-5 = \cancel{0}5 \not> 0$

$(1-3x)$ $f(0) = 1-3(0) = \underline{\underline{1}}$

$f(5/3) = \cancel{3(5/3)-5} = 0$

$1-3x(5/3) = -4 \leq 0$

Let $f^{-1}(0) = x$

$\Rightarrow 0 = f(x)$

$3x-5 = f(x)$	$1-3x = f(x)$
$3x-5=0$	$\Rightarrow 1-3x=0$
$\Rightarrow 3x=5$	$\Rightarrow \cancel{-}3x = \cancel{-}1$
$\Rightarrow x = 5/3 > 0$	$\Rightarrow x = 1/3 \not\leq 0$

$\therefore f^{-1}(0) = 5/3.$

$f^{-1}(5/3)$, let $f^{-1}(5/3) = x$

$\Rightarrow 5/3 = f(x)$

$3x-5 = f(x)$	$1-3x = f(x)$
$\Rightarrow 3x=5 = \frac{5}{3}$	$\Rightarrow 1-3x = 5/3$
	$\Rightarrow 3-9x = 5$
$\Rightarrow 9x-15=5$	$\Rightarrow -9x = 2$
$\Rightarrow 9x = 20$	$\Rightarrow x = \frac{-2}{9}$
$\Rightarrow x = \frac{20}{9}$	

$\therefore f^{-1}(5/3) = \left\{ 20/9 \Rightarrow \frac{-2}{9} \right\}$

x